Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
		(ii)	Use of thermal conductivity equation i.e. $\frac{\Delta Q}{\Delta t} = \frac{0.041\times 72\times 15}{100\times 10^{-3}}$ (1) = 443 [J s^{-1} / W] (1)	2			2	2	
		(iii)	$\frac{\Delta Q}{\Delta t}$ the same through both layers (1) $\frac{0.041(20-\theta)}{100} = \frac{0.035(\theta - 5)}{170}$ seen or boundary temp = 15.0 °C (1) $\frac{\Delta Q}{\Delta t}$ = 148 [W] (1) Accept thermal resistance in series approach i.e. $\frac{15}{\left(\frac{100 \times 10^{-3}}{0.041 \times 72}\right) + \left(\frac{170 \times 10^{-3}}{0.035 \times 72}\right)}$ = 148 [W] > 60% reduction so claim is correct. Allow **ecf** on $\frac{\Delta Q}{\Delta t}$ values(1)			4	4	3	